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CS-499: Computer Science Capstone 2026

Professor Joseph Conlan

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Enhancement Three: Databases

Narrative:

1. Briefly describe the artifact. What is it? When was it created?

The selected artifact in fulfillment of category 3 database enhancement is an Event Planning mobile Android Application that was created in a CS-360: Mobile Architect & Programming course back in March of 2026. The application was designed to provide event planning capabilities for users looking to store upcoming events that contain basic parameters such as the title, description, date, and phone number of the event. Once an event arrives within 7 days of the present day, a notification pops up on the screen, and an SMS message is sent to the phone number parameter for each event. The mobile application has username and password login functionality that has undergone a database enhancement that encrypts and hashes a user's password using a randomly generated salt during account creation by using PBKDF2 with HMAC-SHA512 and stores it in hexadecimal. Once a user inputs a username and password into the input fields, the password is compared to the hashed password stored within the database by using the salt associated with the username stored in the table of the database.

2. Justify the inclusion of the artifact in your ePortfolio. Why did you select this item?

What specific components of the artifact showcase your skills and abilities in software development? How was the artifact improved?

The reason for selecting this artifact for my ePortfolio is because I found the project to be conducive for enhancement in each of the three categories required by the final project of this course. By making the appropriate enhancements in software design & engineering, data structures & algorithms, and databases I am able to effectively fulfill and demonstrate each of the five course outcomes. I also find using Android studio to be intuitive for mobile app development, and I find it convenient to be able to make changes within a seamless IDE such as Android Studio.

A specific component of the artifact that showcases my skills and abilities in software development is the account creation and login verification functions of the application. Prior to enhancement, the username and password data was hardcoded by using plaintext then storing them into the database without applying encryption best practices to ensure data security. As an improvement on the database, I have chosen to implement PBKDF2 with HMAC-SHA512 for hashing all string passwords by using a randomly generated salt and storing both the hashed password and salt into the database. Once a user attempts to login into the application to access the event planner, the entered password is hashed by using the stored salt and is then compared to the hashed password stored in the database and checks whether both match.

3. Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

I believe that I have met all course outcomes that I had planned to meet for the enhancement three. Enhancement three course outcomes demonstrates the ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals (software engineering/design/database) and designing, developing, and delivering professional-quality oral, written, and visual communications that are coherent, technically sound, and appropriately adapted to specific audiences and contexts. These two course outcomes were met once the PBKDF2-HMAC-SHA512 hash was implemented, along with use of industry standard comments written throughout the codebase. I do not have any updates to my outcome-coverage plan thus far.

4. Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

The process of enhancing and modifying my artifact was through extensive research into various cryptographic algorithms and understanding ones that are most effective in protecting login credentials. I then decided to implement the PBKDF2 with HMAC-SHA512 and had to examine a number of example codes online in order to thoroughly understand the logic of hashing and verification with PBKDF2. Then, I replicated the code to function off of password inputs into a field in the main activity, with several instances of bugs that I had to clean up along the way. Having gone through the entire process of merely understanding the theory of cryptography, I

was able to implement a deployable application architecture that taught me about the importance of defensive programming and credential management.